Transparent Synchronous Electrostatic Actuator for Long-Stroke Planar Motion

Takuya Hosobata, *Member, IEEE*, Akio Yamamoto, *Member, IEEE*, and Toshiro Higuchi, *Member, IEEE*

***Abstract*—This paper reports a development of a transparent planar electrostatic actuator, which is useful for constructing interactive human–computer interfaces on flat panel displays. The actuator is a pair of transparent plastic films with strip electrodes in three-phase connections. Half of the electrodes in each film are X electrodes, and the rest are Y electrodes; electrostatic forces excited by applying three-phase voltage to these electrodes drive the actuator at speeds proportional to the excitation frequency. The stator film of the actuator has its X and Y electrodes alternately arranged in a four-by-four checkerboard pattern. The slider is a square film with its side twice that of a single square in the stator, with its electrodes separated by diagonal lines. A geometric analysis shows that this electrode arrangement brings position independence of overlapping areas of electrodes. The position independence gives the actuator a uniform thrust force characteristic required for long-stroke planar motion. Based on this concept, we fabricated a prototype by screen-printing. The prototype could exert approximately 100-mN thrust force with an excitation by a pair of 500-V three-phase voltages, and could travel along straight and curved paths within a 132 mm by 132 mm movable range.**

***Index Terms*—Actuators, electrostatic devices, films, flat panel displays, printed circuits.**

I. Introduction

USER-CENTERED mechatronic systems often require their actuators to be hidden from their users. This applies especially to human–computer interfaces that present motions and/or forces of actuated objects together with visual information. In such interfaces, the actuated objects must blend in with displayed computer graphics, to allow the user to focus on the interaction. Thus, the actuators are not welcomed in the user's sight; they must obscurely play their roles as magic tricks. In this paper, we report a transparent electrostatic actuator shown in Fig. 1, which fulfills such requirement.

One actively studied class of human–computer interface is an active tabletop interface, whose platform is a visual display, e.g., a flat panel display or a projector screen. The interface provides not only the visual information on the platform, but also physical information such as forces and motions, by having real objects actuated on the surface of the platform. For such interfaces, successful implementations of planar magnetic motors have been reported, which include arrays of electromagnets [1], [2], planar induction motors [3], and Sawyer motors [4]. Although not reported for the interface application, planar synchronous motors with moving magnets [5]–[7] are suited for the purpose as well. Other than these planar motors, wire driven actuators [8] and magnetic levitators [9], [10] are also promising. These actuators can present forces and motions in collaboration with computer graphics, by placing them beneath surfaces of visual displays. The surface can also be a plain table in case of using augmented reality technology as in [8]. Although these actuators are sufficiently powerful for the application, their problem is in the difficulty of installation; a large volume is required in the system for the actuators to occupy.

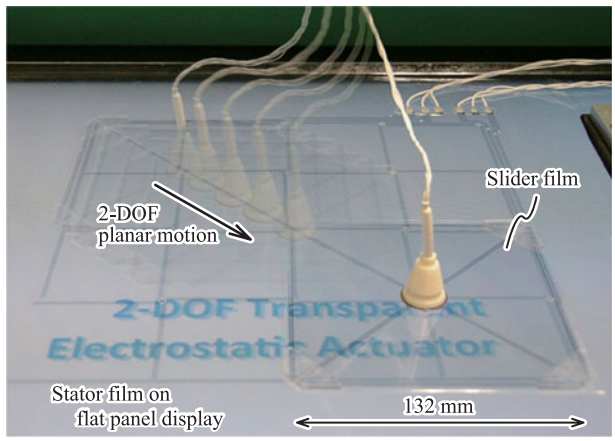


Fig. 1. Transparent SPEA on flat panel display.

Another approach to keep the actuators out from the user's sight is to make the actuators invisible or inconspicuous. This can be done using wire-driven actuators with thin wires [12], surface acoustic wave actuators with glass substrate [11], or electrostatic actuators that are made of transparent films [13]. These actuators can be placed on top of visual displays without blocking the user's vision. Among these actuators, we propose the use of transparent electrostatic actuators, because of their outstanding easiness of installation. The transparent film structure allows us to construct active tabletop interfaces on readily available flat panel displays, just by placing them on the display surface.

The original concept of the transparent electrostatic actuator was proposed by Egawa *et al.* in 1991 [13]. This actuator was a polyethylene terephthalate (PET) film with strip-electrodes on its surface patterned with a transparent conductor (Indium Tin Oxide, ITO). The actuator could convey dielectric sheets on its surface by an electrostatic force excited by voltages applied to its electrodes. The first application of this actuator to a tabletop

Manuscript received March 4, 2014; revised July 2, 2014; accepted August 11, 2014. Date of publication September 24, 2014; date of current version August 12, 2015. Recommended by Technical Editor R. Oboe. This work was supported by Funding Program for Next Generation World-Leading Researchers (#LR013) from JSPS, Japan.

The authors are with the Department of Precision Engineering, School of Engineering, University of Tokyo, Tokyo 113-8654, Japan (e-mail: hosobata@aml.t.u-tokyo.ac.jp; akio@aml.t.u-tokyo.ac.jp; higuchi@aml.t.u-tokyo.ac.jp).

Color versions of one or more of the figures in this paper are available online at http://ieeexplore.ieee.org.

Digital Object Identifier 10.1109/TMECH.2014.2353815

1083-4435 © 2014 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See http://www.ieee.org/publications_standards/publications/rights/index.html for more information.

Authorized licensed use limited to: Southern University of Science and Technology. Downloaded on September 15,2020 at 00:41:38 UTC from IEEE Xplore. Restrictions apply.

interface was reported in [14], but presentable motion by this actuator was limited because the actuator was an asynchronous actuator and had only 1 degree of freedom (DOF). To provide a 2-DOF transparent actuator with improved controllability, we developed the synchronous planar electrostatic actuator (SPEA) [15]. The SPEA was developed based on a well-studied 1-DOF synchronous electrostatic actuator [16], inheriting its open-loop positioning capability. In our previous study, we reported the SPEA's basic concept with the actuator's first prototype [15]. Further, we demonstrated its application in an interactive system on a flat panel display [17].

The contribution of this paper is in providing an analysis method for the SPEA, with emphasis on the geometry of electrode arrangement. The SPEA in [15] was composed of two films: the stator film and the slider film, both of which were patterned with three-phase strip-electrodes for x- and y-directions. The X and Y electrodes were separately arranged in checkerboard patterns, with slider checkerboard half-sized compared to the stator's. This arrangement provided a position independent force characteristic in a broad planar work space which was 132 mm $\times$132 mm. However, there was no theory to backup the effectiveness of such arrangement, because it was chosen empirically. In this paper, we clarify the conditions that must be satisfied by the electrode arrangement. Additionally, based on the derived conditions, we present a novel slider design with improvements in simplicity and rotational stability, which has its electrodes separated by diagonal lines.

Readers should note that the analysis presented in this paper is in a simplified form: quasi-static and two-dimensional (2-D). The analysis assumes that the thrust force characteristic per unit area is given, measured or simulated beforehand. The aim of this simplification is to focus on providing an entry point for future designers of the SPEA in the earliest stage of design, where they must decide where to place the X and Y electrodes. The next step of design: optimization of microscopic structures of these electrodes considering their dynamic movements and excited electric field, is an important and challenging problem, but is out of scope of this paper.

The remainder of this paper is organized as follows. In Section II, we explain the basic structure and principle of the SPEA, and clarify its design problem. Section III provides an analysis on the geometry of electrode arrangement and its effect on the actuator's performance. The analysis enables calculation of the thrust force and the rotational restoring moment of the actuator, and reveals the conditions that must be satisfied to realize a long-stroke planar motion. Based on this analysis, we designed and fabricated a prototype, which is described in Section IV. The prototype is made of transparent materials, and the key fabrication technique is screen-printing. In this prototype, some ideal conditions were not satisfied due to practical limitations. The effect of this on the performance of the actuator is also described in this section. In Section V, we report the results of performance evaluation. The results include planar motion along straight and curved paths, position dependence of thrust force, and the maximum acceleration with payload. Section VI is the discussion on improvements necessary for the SPEA. Although this study has improved the actuator from the previous prototype, it is still in the early stage of development; the actuator is not ready for immediate application in practice. Works that need to be done toward practical implementation of this actuator in human–computer interface will be discussed in this section. These works include optimization of electrode structure, removal of wires from the slider, searching for suitable guides, and assurance of safety.

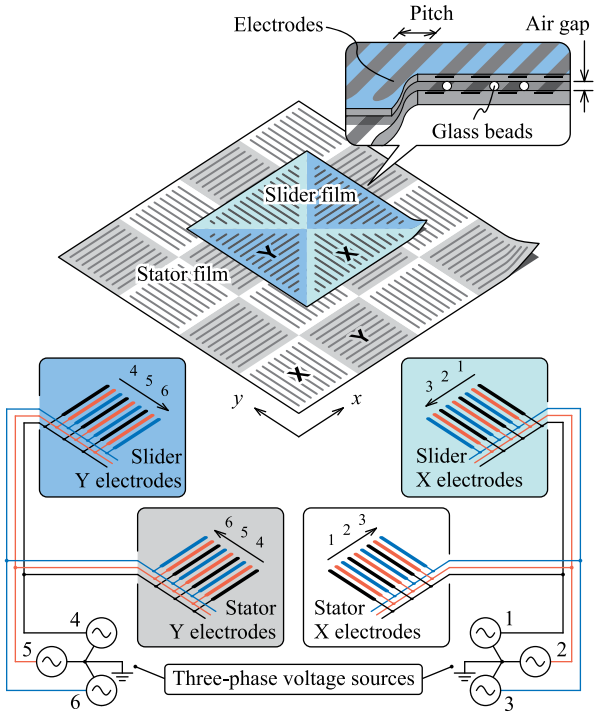


Fig. 2. Schematic of SPEA with checkerboard-patterned stator and cross slider.

II. SPEA

A. Structure of SPEA

Fig. 2 shows the schematic of the SPEA. The actuator is an overlaid pair of films. The larger film fixed at the bottom is the stator and the smaller film placed on top is the slider. Between these two films are glass beads maintaining an air gap.

Each film has three-phase strip-electrodes for driving in two directions x and y, which are, respectively, referred to as the X and Y electrodes. The X electrodes in the stator and the slider are connected to the same three-phase voltage source but in opposite phase sequences, and so are the Y electrodes. For further reference, phase numbers one to three are given to the X electrodes and four to six are given to the Y electrodes.

In each film, the X and Y electrodes are arranged in separate regions. In the stator, the X and Y electrodes appear alternately in squares of a checkerboard pattern. This arrangement was determined so that the X and Y electrodes have the same areas in the stator. Whereas in the slider, the X and Y electrodes are sectioned by diagonal lines. This slider design differs from that in our previous study [15], improved both in simplicity and stability. This slider—further referred to as the cross slider—can operate with a constant thrust force characteristic over the

Authorized licensed use limited to: Southern University of Science and Technology. Downloaded on September 15,2020 at 00:41:38 UTC from IEEE Xplore. Restrictions apply.

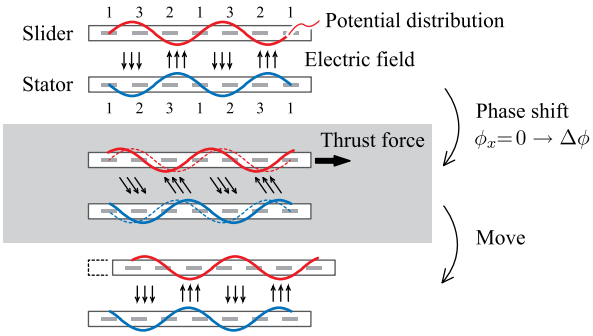


Fig. 3. Principle of synchronous electrostatic actuator.

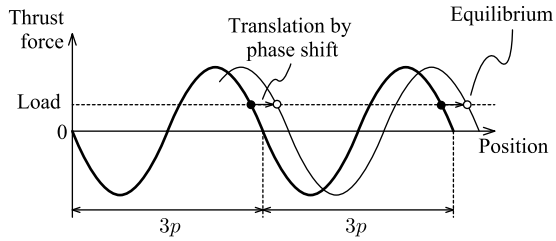


Fig. 4. Idealized thrust force characteristic of synchronous electrostatic actuator.

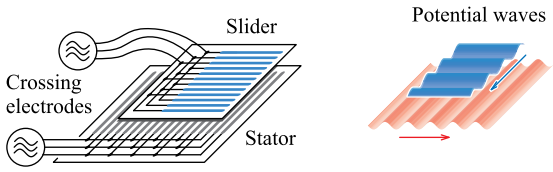


Fig. 5. Potential waves formed on electrodes arranged in different directions.

whole work space on the stator. Details of how the cross slider was designed will be described in Section III.

B. Principle of 1-DOF Actuator

For simplicity, we start the explanation on the operating principle of the actuator using an 1-DOF model of synchronous electrostatic actuator in Fig 3. The explanation is kept simple, as the principle is well described in preceding studies [16], [18]. First, we define the three-phase voltage applied to the electrodes one to three as

$$[v_1\ v_2\ v_3] = V\left[\sin\phi_x\ \ \sin\left(\phi_x - 2\pi/3\right)\ \ \sin\left(\phi_x - 4\pi/3\right)\right] \tag{1}$$

where v_k is the temporal voltage applied to the electrode number k, V is the voltage amplitude, and ϕ_x is the temporal phase. These voltages form a potential distribution on the facing surfaces of the two films, which are spatially periodic with a period of $3p$, where p is the electrode pitch.

At first, let us assume that ϕ_x is kept constant at $\phi_x = 0$. At this condition, the slider rests at a position where the two potential distributions are aligned with horizontal symmetry, because the symmetric potential distributions excite vertical electric field. Next, say we instantly shift the phase ϕ_x by a small amount of $\Delta\phi$. The shifting of phase causes the stator potential to shift forward by $(3p/2\pi)\Delta\phi$. At the same time, the slider potential is shifted backward by the same amount, because the phase sequence of the voltages are opposite from the stator. As a result, the two potential distributions are spatially displaced by $(3p/\pi)\Delta\phi$, which gives rise to a tangential component in the electric field. The tangential field exerts a thrust force on the slider, and forces it to advance. Finally, the slider rests at a new equilibrium position, which is $(3p/\pi)\Delta\phi$ shifted from the previous one. In actual driving, we apply ac voltage; thus, the phase ϕ_x is continuously shifted and so is the slider's equilibrium position. In one cycle of the voltage: when ϕ_x is shifted by 2π, the slider advances by $6\ p$. Therefore, the speed of the slider excited by a three-phase voltage with frequency f is $6\ pf$, which is called the synchronous speed.

According to the model proposed in [18], an idealized thrust force F_x of the 1-DOF actuator is expressed as

$$F_x = -K_F V^2 \sin\left(2\pi x_c/3p - 2\phi_x\right) \tag{2}$$

where K_F is a thrust force constant in units of $\mathrm{N/V^2}$ and x_c is the slider's position. The term "idealized" indicates that this expression excludes a thrust force ripple. Analysis of the ripple is out of scope of this paper, but will be explained in Section V, as they appear in experimental results.

The thrust force characteristic is illustrated in Fig. 4. The actuator is able to operate only when the applied load is less than the maximum force $K_F V^2$, and under this condition, the position of the slider is controllable by gradual shifting of the phase of the voltage because the equilibrium position, where the thrust force balances with the load, is also shifted. There are many equilibrium positions, because the electrode structure is periodic by a period of $3\,p$. Thus, when the slider is exposed to an instantaneous and excessive disturbance, it jumps to another equilibrium position; this we call step-out of the slider.

C. Thrust Force and Overlapping Area

In explaining the principle of 1-DOF synchronous actuator, we implicitly assumed that the overlapping area of the electrodes does not change. For the SPEA, the overlapping area may change with the position of the slider. From Fig. 1, we can see that not all slider electrodes are facing their parallel counterpart in the stator; some portions of X electrodes are overlapping with Y electrodes and *vice versa*. In these regions, the potential waves formed on the stator and the slider travel in different directions as shown in Fig. 5, and these waves do not contribute in producing thrust force. Thus, only the regions containing the same-direction electrodes are effective. With this considered, we define the thrust forces of the SPEA in x- and y-directions as

$$\begin{cases} F_x = -k_F A_{ex} V^2 \sin\left(2\pi x_c/3p - 2\phi_x\right) \\ F_y = -k_F A_{ey} V^2 \sin\left(2\pi y_c/3p - 2\phi_y\right) \end{cases} \tag{3}$$

where k_F is the thrust force constant per unit area in units of $\left(\mathrm{N/m^2}\right)/\mathrm{V^2}$, A_{ex} and A_{ey} are effective overlapping areas of X

Authorized licensed use limited to: Southern University of Science and Technology. Downloaded on September 15,2020 at 00:41:38 UTC from IEEE Xplore. Restrictions apply.

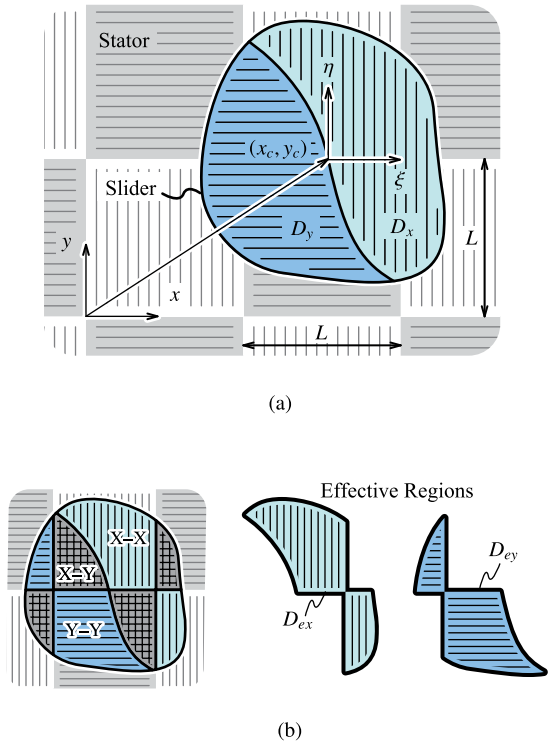


Fig. 6. Definitions of symbols for calculation of effective overlapping areas. (a) Definition of stator and slider coordinates. (b) Effective overlapping regions.

and Y electrodes, y_c is the slider position in y direction, and ϕ_y is the phase of the three-phase voltage for Y electrodes, which is defined as

$$[v_4\ v_5\ v_6] = V\left[\sin\phi_y\ \ \sin\left(\phi_y - 2\pi/3\right)\ \ \sin\left(\phi_y - 4\pi/3\right)\right]. \tag{4}$$

In order to realize a planar actuator that has uniform characteristic at all positions in its stator plane, the effective overlapping areas A_{ex} and A_{ey} must be independent of the slider position. However, with an inadequately designed slider, the effective overlapping areas would change greatly; in the worst case, the effective areas are completely lost at certain positions. Whereas, the cross slider we introduce in this paper has position independent overlapping areas, because its electrode arrangement satisfies the conditions which are described in the following section.

III. Geometric Analysis

In this section, we derive the conditions that are to be satisfied by a slider for an SPEA, in order to have constant overlapping areas of X and Y electrodes, and hence the position independent thrust force characteristics. Further, we extend our analysis to include yaw-moment, which is related to the passive rotational stability of the SPEA.

A. Definitions

As shown in Fig. 6(a), we assume that the stator has its X and Y electrodes sectioned into squares whose four sides have length L, arranged in an infinite checkerboard pattern. The slider is assumed to have its X and Y electrodes in regions of arbitrary shape. Stator coordinates and slider coordinates are, respectively, defined as (x, y) and (ξ, η). The slider coordinates have its origin at $(x, y) = (x_c, y_c)$, which is fixed to the slider. The region including X electrodes of the slider is represented by D_x, and the same for the Y electrodes is represented by D_y. We assume that these regions are defined in slider coordinates. Fig. 6(b) shows the effective regions for this slider at a certain position. Symbols D_{ex} and D_{ey} are used to represent these regions.

B. Calculation of Effective Overlapping Area

In order for the effective overlapping area of X electrodes to be independent of the slider position (x_c, y_c), it must satisfy

$$\frac{\partial A_{ex}}{\partial x_c} = 0, \quad \frac{\partial A_{ex}}{\partial y_c} = 0. \tag{5}$$

Thus, the first goal of analysis is to express the effective overlapping area as a function of the slider position. The effective overlapping area is calculated by the integration

$$A_{ex} = \iint_{D_{ex}} d\xi d\eta \tag{6}$$

but this calculation is difficult because D_{ex} changes its shape with the slider position. To simplify the integration, we define a checkerboard function $f_{cb}(x, y)$ as a 2-D periodic function with a period of $2L$, whose values in one period is expressed as

$$f_{cb}\left(x, y\right) = \begin{cases} 1 & (0 \le x < L, \quad 0 \le y < L) \\ 0 & (-L \le x < 0, \quad 0 \le y < L) \\ 1 & (-L \le x < 0, \quad -L \le y < 0) \\ 0 & (0 \le x < L, \quad -L \le y < 0). \end{cases} \tag{7}$$

This function takes a value of 1 when (x, y) is within the X electrode region of the stator, and 0 when it is within the Y electrode region of the stator. Using this function, (6) is rewritten as

$$A_{ex} = \iint_{D_x} f_{cb}\left(\xi + x_c, \eta + y_c\right) d\xi d\eta. \tag{8}$$

Note that the region of integration has been changed from D_{ex} to D_x. This is because multiplying of f_{cb} eliminates infinitesimal areas $d\xi d\eta$ within ineffective regions from the integrated area. The integration in (8) is still too complicated for analytical calculation, because the function f_{cb} is expressed in inequity notations. Thus, for further simplification, we apply 2-D Fourier expansion to f_{cb}, which yields

$$f_{cb}\left(x, y\right) = \frac{1}{2} + \frac{8}{\pi^2} \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \frac{\sin \mu\alpha x \sin \nu\alpha y}{\mu\nu} \tag{9}$$

where $\mu = 2m - 1$, $\nu = 2n - 1$, and $\alpha = \pi/L$.

Authorized licensed use limited to: Southern University of Science and Technology. Downloaded on September 15,2020 at 00:41:38 UTC from IEEE Xplore. Restrictions apply.

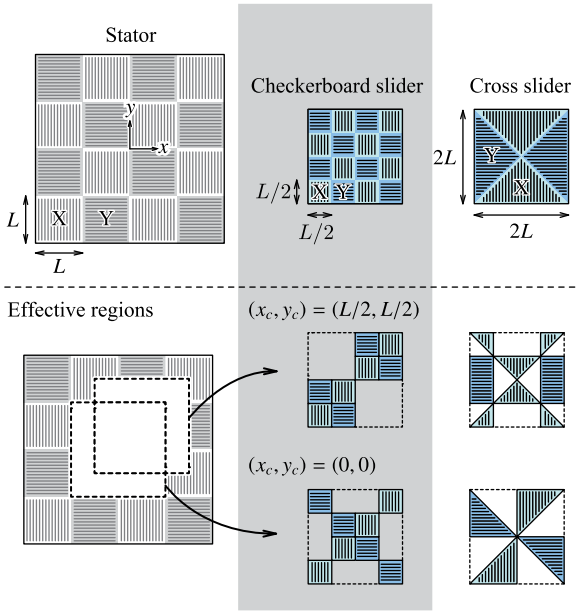


Fig. 7. Two sliders with position independent overlapping areas.

From (5), (8), and (9), the conditions of D_x for position independence of A_{ex} are derived as

$$\begin{cases} \int\int_{D_x} \cos\mu\alpha\xi \cos\nu\alpha\eta d\xi d\eta = 0 \\ \int\int_{D_x} \cos\mu\alpha\xi \sin\nu\alpha\eta d\xi d\eta = 0 \\ \int\int_{D_x} \sin\mu\alpha\xi \cos\nu\alpha\eta d\xi d\eta = 0 \\ \int\int_{D_x} \sin\mu\alpha\xi \sin\nu\alpha\eta d\xi d\eta = 0. \end{cases} \tag{10}$$

This indicates that the region D_x must be chosen so that all odd harmonics of the fundamental space frequency $2L$ become null when integrated. The conditions of D_y are derived in a similar manner; the only difference is that f_{cb} is replaced by $1 - f_{cb}$, and the results are the same as (10) with D_x replaced by D_y.

C. Sliders With Position Independent Overlapping Areas

There are an infinite number of shapes for D_x and D_y that satisfy (10), but in this paper we focus on the two simple solutions we found, which are shown in Fig. 7. The slider on the left is a checkerboard slider we reported in our previous study [15]. The slider has its X and Y electrodes in a checkerboard pattern with sides of squares half that of the stator. The slider on the right is the cross slider with the X and Y electrodes separated by diagonal lines, which we introduce in this study.

For both of these sliders, calculation of the overlapping areas yields $A_{ex} = A_{ey} = L^2$ (const.); thus, they have equivalent thrust force characteristics, independent of the slider position. However, the cross slider is better in two other ways. First, it is simpler. As will be shown in Section IV, increase in the number of separate regions complicates the patterning of electrodes. The cross slider has only four separate regions, which should be the minimum number considering the symmetry for rotational stability. Second, restoring yaw moment of the cross slider is 50% larger than the checkerboard slider, which indicates an improvement in the rotational stability. In the next section, we show how the yaw moments are calculated.

D. Restoring Yaw Moment

The 1-DOF synchronous electrostatic actuator is known to be rotationally stable [19], and so is the SPEA. Provided that the rotational displacement is small enough, a yaw moment acts on the slider to realign the electrodes. To calculate for this yaw moment, we first think about infinitesimal effective area of X electrodes, dA_{ex}, at position (ξ, η) in slider coordinates. A counter-clockwise rotation by a small angle θ displaces the electrodes in this small area by $-\eta\theta$ in the x-direction. Similarly, for Y electrodes, the displacement is $\xi\theta$ in y-direction. From (3), with an assumption that the slider rests at its equilibrium position at an initial state, the thrust forces exerted on the small areas dA_{ex} and dA_{ey} are expressed in linearized forms as

$$\begin{cases} dF_x \approx (2\pi/3p)\, k_F V^2 \eta\theta dA_{ex} \\ dF_y \approx -(2\pi/3p)\, k_F V^2 \xi\theta dA_{ey}. \end{cases} \tag{11}$$

Based on these forces, the restoring yaw moment M_z is calculated as

$$M_z = -\iint_{D_x} \eta dF_x + \iint_{D_y} \xi dF_y = -K_M V^2 \theta \tag{12}$$

where K_M is the yaw moment coefficient in units of $(\mathrm{N}\cdot\mathrm{m/rad})/\mathrm{V}^2$, which is calculated as

$$K_M = \frac{2\pi}{3p} k_F \left(\iint_{D_x} \eta^2 dA_{ex} + \iint_{D_y} \xi^2 dA_{ey} \right). \tag{13}$$

To simplify this integration, we can also use the function f_{cb} in (9), which yields

$$K_M = \frac{2\pi}{3p} k_F \left\{ \iint_{D_x} \eta^2 f_{cb}\left(\xi + x_c, \eta + y_c\right) d\xi d\eta + \iint_{D_y} \xi^2 \left(1 - f_{cb}\left(\xi + x_c, \eta + y_c\right)\right) d\xi d\eta \right\}. \tag{14}$$

By (14), we can calculate the yaw moment coefficients of the two sliders. The coefficient of the checkerboard slider is

$$K_M = \frac{4\pi}{9p} L^4 k_F \ (\text{const.}) \tag{15}$$

and the coefficient of the cross slider is

$$K_M = \frac{2\pi}{3p} L^4 k_F \left(1 - \frac{64}{\pi^4} \sum_{m \neq n} \frac{\sin\mu\alpha x_c \sin\nu\alpha y_c}{\mu\nu(\mu^2 - \nu^2)} \right) \tag{16}$$

where μ, ν, and α are the same as those defined for f_{cb} in (9). The second term in (16) indicates that the yaw moment coefficient of the cross slider is position dependent. However, as shown in Fig. 8, this position dependent change is only $\pm 4\%$ of the average value, which is $(2\pi k_F/3p)\, L^4$. This average value is

Authorized licensed use limited to: Southern University of Science and Technology. Downloaded on September 15,2020 at 00:41:38 UTC from IEEE Xplore. Restrictions apply.

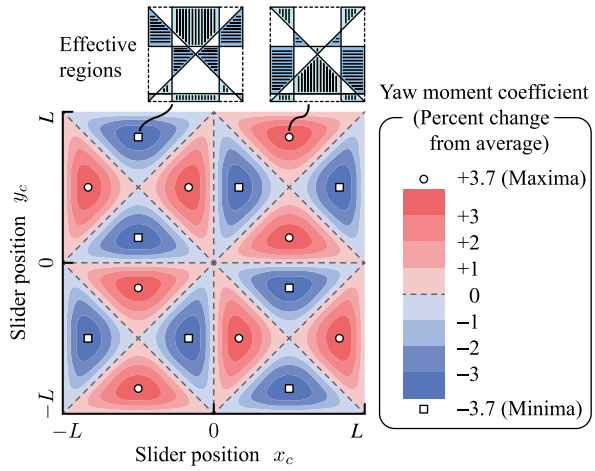


Fig. 8. Position dependent change of yaw-moment coefficient.

50% larger compared to that of the checkerboard slider. This difference is apparent in the shapes of effective overlapping areas in Fig. 7; the cross slider has more effective electrodes toward the outside of the slider. In most applications, the cross slider should be preferred over the checkerboard slider, even with the position dependence of the yaw moment coefficient, as larger yaw moment coefficient brings a great improvement in the stability of the actuator.

IV. Prototype of Transparent Actuator

This section describes the transparent stator and slider films, designed and fabricated based on the analysis in the previous section. The key process is screen-printing, which is recently drawing attention as a cost-effective method of fabrication for flexible printed circuits. One problem in design arises from practical limitations in screen-printing: bus connections of strip electrodes require spacings between electrode regions. These spacings cause the ineffective overlapping area to increase, which was not included in the analysis. The effect of these spacings are analyzed in the later part of this section.

A. Checkerboard-Patterned Stator

Fig. 9 shows the transparent stator film. The X and Y electrodes are strip electrodes with 400-μm pitch, patterned on an ITO-deposited PET film 125 μm thick (HNC125-AB150, Teijin) by lithography and etching. These electrodes are separated in 16 regions of a four-by-four checkerboard pattern with the sides of each square 66 mm. These electrodes were connected in three-phase patterns by the following screen-printing process.

The prepared ITO pattern has two of the three-phase electrodes in each square connected to the bus lines along its side, but the electrodes for one of the phases are electrically floating. On top of this film, we print a transparent insulating layer using thermosetting plastic paste (FR-1T-NSD9, Asahi Chemical Research Laboratory). The thickness of this insulating layer is approximately 20 μm. This layer has holes, which leaves tips of the floating electrodes uncovered. To establish vias through these holes, we printed bus lines on top of this insulating layer, using a conductive polymer PEDOT:PSS (Orgacon EL-P5015, Agfa-Gavert Group), which is an abbreviation for poly(3,4-ethylenedioxythiophene):poly(4-styrenesulfonate). We chose PEDOT:PSS for good printability and transparency; the bus lines appear blue in the photograph, but they are less prominent on flat panel displays as was shown in Fig. 1. Finally, the whole surface of the film was covered with another insulating layer with the same material and thickness as the previous one.

The most difficult part in design was to have the bus lines for X and Y crossed. To realize the crossing, we designed the bus-intersection as shown in the lower-right corner of Fig. 9. The vias for this intersection was filled with silver paste. This was necessary to increase the withstanding current of the vias; because the actuator is capacitive, inrush currents flow through these vias when voltages are suddenly applied.

B. Cross Slider

Fig. 10 shows the prototyped cross slider. Unlike the stator, all electrodes including the three-phase electrodes are screen-printed using PEDOT:PSS, on a blank sheet of PET that is 50 μm thick. Except for this difference in materials, the slider shares the same fabrication process as the stator.

The slider has terminals at its center, to which cables from the voltage sources are connected. This is to reduce the rotational disturbance from the cables. The slider has plastic weights at its four corners; without these weights the film is curled, because of the residual stress in insulating layers. The film alone weighs 2.4 g, and the whole slider including these weights and the cable holder at the center weighs 10.9 g.

C. Spacings Between Regions

It is apparent from the designs of these two films that there are extra spacings between the X and Y regions for bus lines and vias, that were not included in the analysis. These spacings cause the overlapping areas to change with the slider position, which has significant effect on the thrust force. Fig. 11 shows these spacings in the actual films. In stator, there are spacings 9 mm wide between each of the squares. Whereas in the slider, there are 3-mm spacings between electrode regions.

Shown in Fig. 12 is the position dependent change of the effective overlapping area of the X electrodes, numerically calculated with the spacings taken into account. The same result for the Y electrodes are not shown, because it is obvious from symmetry; rotating the whole graph around the origin by $+90^\circ$ gives the result. The graph shows the change of overlapping area from its average value, which was 69% of the area of a single square in the stator. The average value indicates that the thrust force is reduced by 30% compared to an ideal case without spacings. Relative to this average value, the effective overlapping area changes by almost ± 10 %; the force in x-direction is weaker in the first and the third quadrant than in the second and the fourth, and the opposite applies to the force in y-direction.

To approach the ideals and lessen the position dependence, the electrode patterns must be optimized to reduce the spacings,

Authorized licensed use limited to: Southern University of Science and Technology. Downloaded on September 15,2020 at 00:41:38 UTC from IEEE Xplore. Restrictions apply.

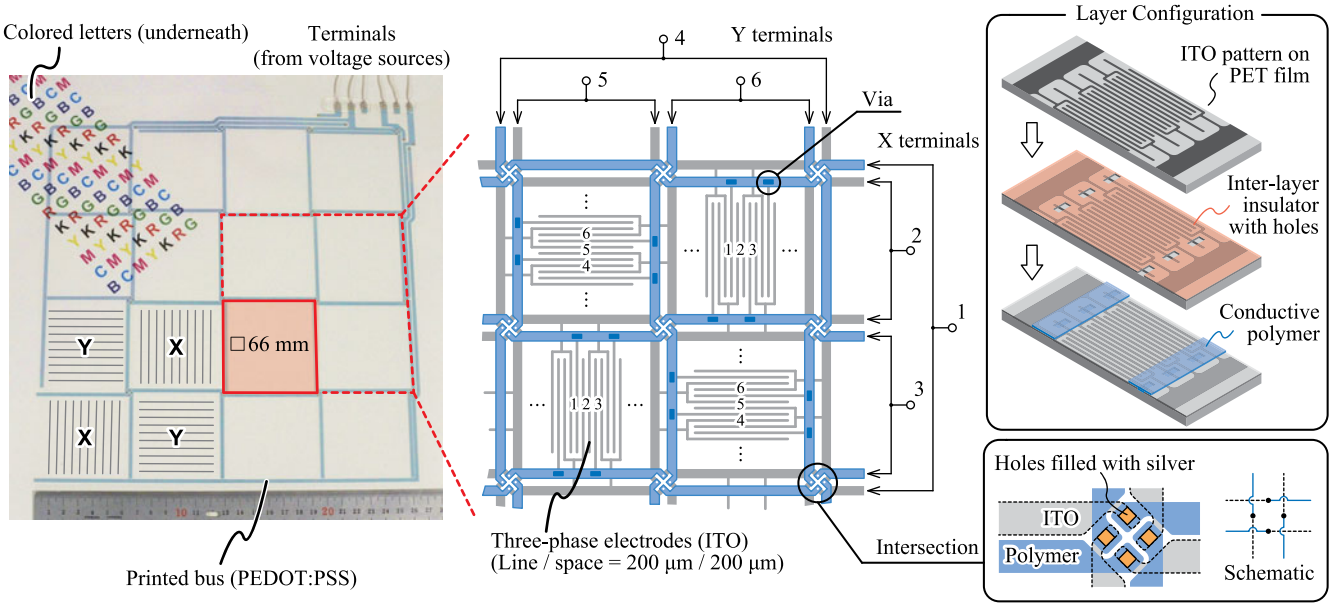


Fig. 9. Transparent stator film with electrodes arranged in a checkerboard pattern.

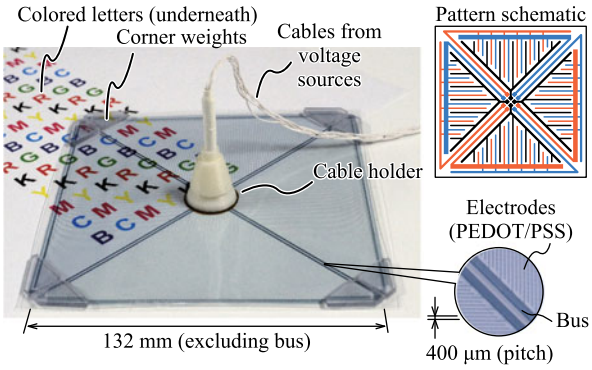


Fig. 10. Prototype of the cross slider.

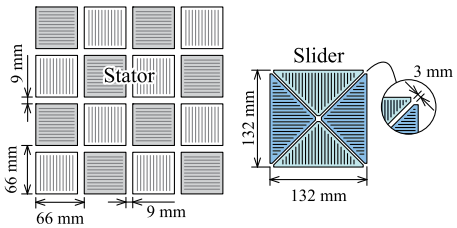


Fig. 11. Spacings between electrode regions for phase bus lines and vias.

with consideration of practical limitations in fabrication process. However, as the experimental results show in the next section, the prototype was able to perform various planar motions even with this position dependence.

V. Experimental Results

This section describes the results of the performance evaluation of the prototyped SPEA. The results include 2-D motion trajectories, thrust forces at various positions, and the maximum

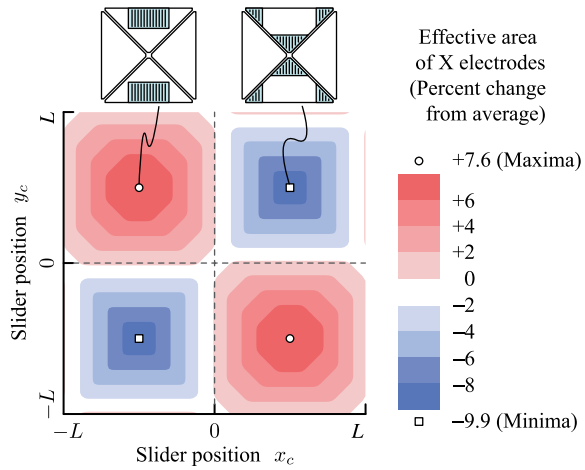


Fig. 12. Position dependent change of X electrodes' effective overlapping area. The symbol L indicates the length of squares in the checkerboard-patterned stator. The average value is 69% of L^2.

accelerations with payload. Note that thrust force ripples appear in these results, which is a nonideal characteristic excluded from the explanation in Section II for simplicity. The explanation on the ripple characteristic will be provided as it appears in the results.

In all experiments, the amplitude of the three-phase voltages applied to both X and Y electrodes were fixed at 500 V_{0-P}. These voltages were generated by amplifying the D/A-converted signals from DSP (DS1104, dSpace) using six high-voltage amplifiers (HVA4321, NF Co.). Also common in all experiments, glass beads with a diameter of 100 μm were scattered between the two films for lubrication and gap maintenance.

A. Measurement of 2-D Motion Trajectory

To show that the developed cross slider is capable of operating in the whole work space on the stator: a square space of

Authorized licensed use limited to: Southern University of Science and Technology. Downloaded on September 15,2020 at 00:41:38 UTC from IEEE Xplore. Restrictions apply.

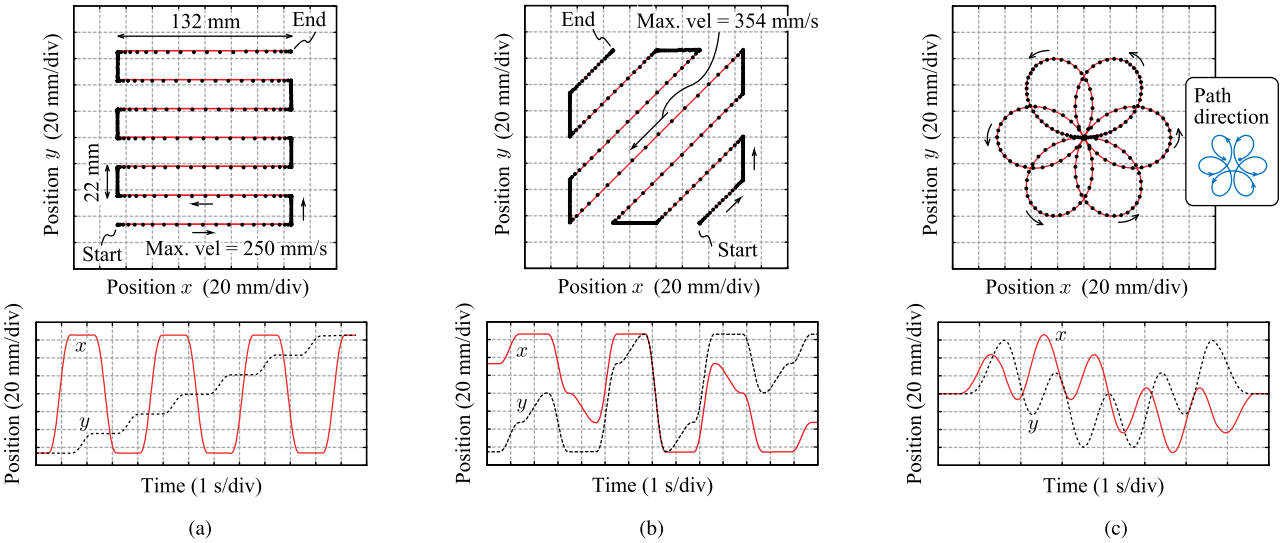


Fig. 13. 2-D open-loop trajectories of synchronous electrostatic planar actuator recorded with a motion tracking camera. The sampling interval for x–y plot is 40 ms, and for t–xy plot is 8 ms. (a) Linear motions in x direction, (b) Linear motions in y direction, (c) Motion along a rose curve.

132 mm by 132 mm, its open-loop motion trajectories along three different paths were recorded. For recording, we used motion tracking camera (VW-6000, Keyence) and the resolution of measurement was 646 μ m/pixel. Fig. 13 shows the results in x–y and t–xy forms. Voltage patterns to realize these motions were determined as follows.

For horizontal straight paths in Fig. 13(a) and diagonal paths in Fig. 13(b), we applied a simple point-to-point trajectory with trapezoidal acceleration/deceleration profile, in which the velocity reaches its maximum at midpoint of the path. For the horizontal paths the maximum velocity and acceleration were set to 250 mm/s and 0.5 $\mathrm{m/s^2}$, and those for the longest diagonal path were set to $\sqrt{2}$ times these values. At first, these point-to-point reference trajectories were expressed as a function of time: $(x_r(t), y_r(t))$. Then, assuming no load in (3), we changed the phases of the three-phase voltages as

$$\phi_x(t) = (\pi/3p)\, x_r(t), \quad \phi_y(t) = (\pi/3p)\, y_r(t). \tag{17}$$

The curved path in Fig. 13(c) is a rose curve represented by a parametric equation with a parameter $0 \le s \le 1$

$$\begin{cases} x_r(s) = R\sin(6\pi s)\sin(4\pi s) \\ y_r(s) = R\sin(6\pi s)\cos(4\pi s) \end{cases} \tag{18}$$

where $R = 66$ mm. For this curved path, we first calculated $\ell(s)$, the length of the path at s from $s = 0$, as

$$\ell(s) = \int_0^s \sqrt{(dx_r/ds)^2 + (dy_r/ds)^2}\, ds. \tag{19}$$

Then, we designed $s(t)$ considering the tangential acceleration $d^2\ell/dt^2$, in a way that the slider would once decelerate and then accelerate at sharp corners. Finally, the designed $s(t)$ was substituted into (18) to obtain the time-series path, which was then substituted into (17) to calculate the phases.

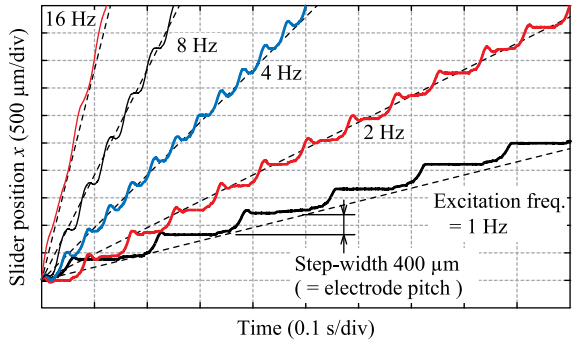


Fig. 14. Step-wise motion of SPEA, driven in x-direction by constant frequency excitation. Dotted lines show the ideal motion.

These motion trajectories indicate the cross slider's capability to travel along arbitrary paths in the whole stator plane. However, because of the rough measurement resolution, which is larger than electrode pitch 400 μm, the results lack the information on microscopic behavior of the actuator. To supplement this, we measured the position of the slider traveling in x-direction, using laser displacement sensor (LC-2400, Keyence). The result is shown in Fig. 14. In this experiment, a three-phase voltage with constant frequency is applied to the X electrodes.

In this result, we can clearly observe a step-wise motion with a step-width equal to the electrode pitch. This step-wise motion comes from the thrust force ripple that has been known to exist also in 1-DOF synchronous electrostatic actuator [16]. Overshoots are observed in the step-wise motion, especially for 2- and 4-Hz excitation. This dynamic behavior is also unexplainable with the idealized model used in this study, because the model does not take friction and damping into consideration. The step-wise motion becomes less prominent as it becomes faster. Likewise, for the motions in Fig. 13, the step-wise motion was visually apparent only in starting and stopping, and otherwise it appeared to be smooth.

Authorized licensed use limited to: Southern University of Science and Technology. Downloaded on September 15,2020 at 00:41:38 UTC from IEEE Xplore. Restrictions apply.

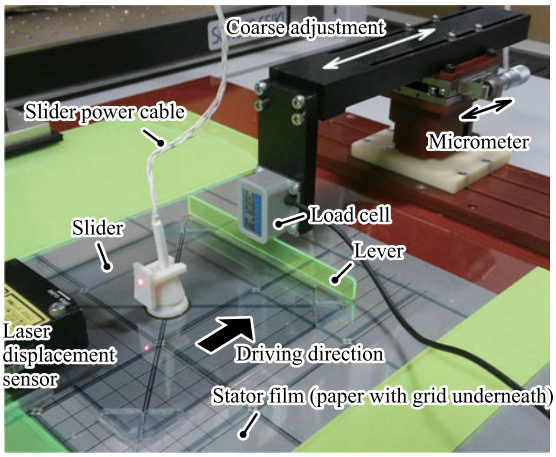


Fig. 15. Experimental setup for thrust force measurement.

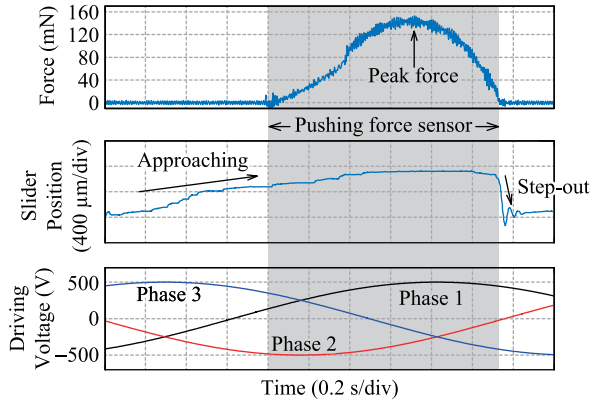


Fig. 16. Example of data obtained in one force measurement.

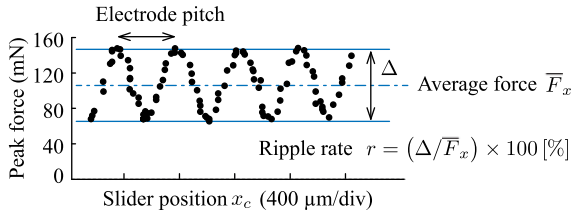


Fig. 17. Thrust force ripple of SPEA.

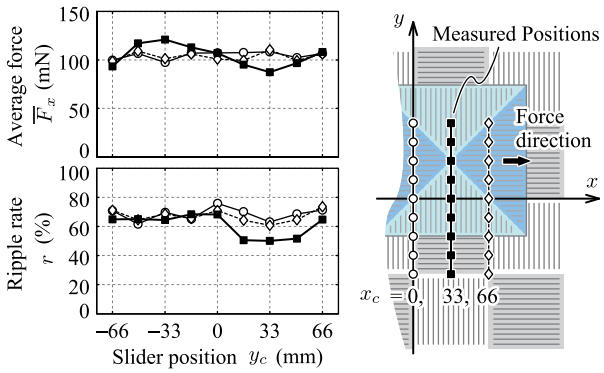


Fig. 18. Average thrust force and ripple rate measured at different slider positions.

B. Measurement of Thrust Force

Thrust forces of the actuator at different slider positions were measured using the setup shown in Fig. 15. In this setup, the thrust force and the slider position were measured simultaneously, by a load cell (LVS-200GA, Kyowa Electronic Instruments) and a laser displacement sensor (LC-2400, Keyence), respectively. To align the slider at designated measuring position, a sheet of paper with a grid was placed beneath the transparent stator film. The measurement was conducted only for x-direction, because the result also explains the force characteristic for y-direction, considering the symmetry.

In this experiment, a 0.5-Hz three-phase voltage was applied to the X electrodes to press the slider's corner weights against a lever fixed to the load cell. Meanwhile, the slider was guided by the force from Y electrodes. In theory, the guiding is possible by applying dc voltages to the Y electrodes, but in practice it is not, because the thrust force of synchronous electrostatic actuators are known to become unstable after application of dc voltages over a long period of time [20]. To avoid this instability, the voltages to the Y electrodes were set as

$$[v_4\ v_5\ v_6] = V \sin 2\pi f_m t \left[0\ \ -\sqrt{3}/2\ \sqrt{3}/2\right] \tag{20}$$

in accordance with the method proposed in [20]. The frequency f_m was set to 50 Hz.

Fig. 16 shows an example of data obtained by this measurement, which is interpreted as follows. First, as we shift the phase of the X voltage, the slider advances and touches the lever of the load cell. Then, as we shift the phase further, the measured force begin to rise, reaches its peak, and then falls until the slider steps out. The force curve measured during this process corresponds to the positive half of the force curve that was shown in Fig. 4. Thus, the peak of this force curve represents the maximum force of the actuator.

Fig. 17 shows the peak forces measured near $(x_c, y_c) = (0, 0)$ with position of the load cell slightly changed in each measurement, using a micrometer. Each data point in the graph indicates a peak force recorded in a measurement just like in Fig. 16. From this result, we can clearly observe the force ripple: a periodic change of the peak force with a period equal to the electrode pitch. For a 1-DOF actuator, this ripple can be reduced by the skewing of electrodes, that is, patterning of the electrodes in the stator and the slider with a slight angular displacement [19], but its applicability to the SPEA is yet to be considered. This "local" position dependence of thrust force must be distinguished from the "global" position dependence resulting from the change of effective overlapping area. Thus, we defined a ripple rate r with respect to the average force $\overline{F}_x$ as shown in the figure.

We measured the thrust forces within ± 0.6 mm of 27 different positions, along three lines $x_c = 0, 33, 66$ mm, and the results are shown in Fig. 18. As expected from the calculation in Fig. 12, the results clearly indicate that the thrust force is minimized in the first quadrant at $(x_c, y_c) = (33, 33)$ mm and maximized in the fourth quadrant at $(x_c, y_c) = (33, -33)$. The position

Authorized licensed use limited to: Southern University of Science and Technology. Downloaded on September 15,2020 at 00:41:38 UTC from IEEE Xplore. Restrictions apply.

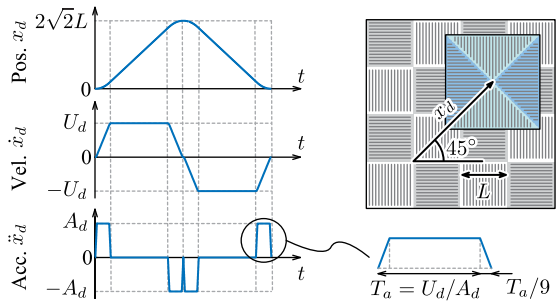


Fig. 19. Diagonal motion trajectory for measuring maximum withstanding acceleration.

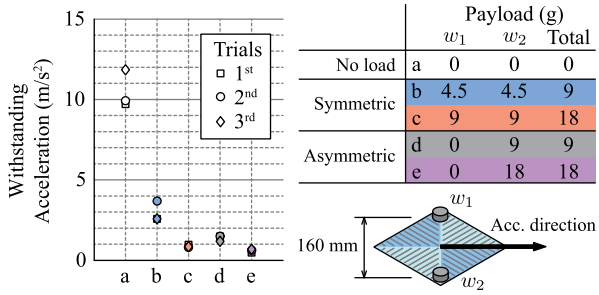


Fig. 20. Maximum withstanding accelerations with different loading conditions. The experiment was conducted three times.

dependent change was approximately $\pm 17\%$, which was larger than the expectation.

C. Maximum Acceleration With Payload

It is important to clarify the payload capacity of the SPEA, because the actuator is developed for tabletop interface applications, in which the slider must carry various objects for presentation. Although we expect that the main interest of the users of the actuator is in the maximum speed with different payloads, we chose the maximum acceleration to represent the actuator's payload capacity, because the maximum speed is a rather indirect measure for this particular actuator due to the stroke limitation; the slider reaches its maximum speed when it uses full stroke to accelerate and decelerate, which means that the speed limitation comes from the allowed acceleration and the given stroke.

We measured the maximum acceleration that the slider withstands, while traveling back-and-forth on a diagonal path shown in Fig. 19. The maximum speed during the motion was fixed at 50 mm/s, and the acceleration/deceleration time was adjusted to reach this speed with designated acceleration. When the slider could come back to its starting position without stepping out, the motion was counted as successful. We searched for the maximum withstanding acceleration by bisecting between successful and unsuccessful conditions.

Fig. 20 shows the results under five different loading conditions. First, in condition a: with no load, the slider could withstand an acceleration of about 10 m/s^2. Then, we put weights on the two corners of the slider in a symmetric manner (conditions b and c). From the graph, we can observe that data for conditions a and b show lower repeatability than the rest. We believe that this comes from the instability in friction. The frictional condition may have varied in each trial, due to the change in the distribution of glass beads caused by the rapid movement of the slider. Readers should note that the results of this measurement include such data variation.

The reduction of withstanding acceleration by these weights were more than expected; expected accelerations from the increase of weight were 5.5 and 3.8 m/s^2, respectively, for addition of 9 and 18 g weights, but the actual values were 2.9 and 0.9 m/s^2 . This discrepancy may be caused by the increase of friction due to the additional weight. Finally, we applied a biased load by placing the weights only on one corner (conditions d and e). The slider could also operate with such biased loads, but the withstanding accelerations were further decreased, almost by half compared to the symmetric loading with the same weights.

In practice, accelerations should be set much lower than these critical values to have enough margin for unexpected disturbances. For presentation of stable motions like those in Fig. 13, a recommendation based on our experience is to assume a safety factor of 10 to 20, with respect to these critical accelerations.

VI. Discussion

Although some practical problems in patterning remain to be resolved, development of the SPEA as a whole was successful, as its capability to operate with a long-stroke was proved in experiments. The thrust force of the actuator is far less than those of magnetic planar motors, but sufficient to present various planar motions of the object it carries, on flat panel displays. We expect that the analysis in this study promotes future ideas in electrode design, which further improves the performance of the actuator.

Toward practical application, however, there are four major challenges aside from the electrode geometry. First, microscopic structure of the electrodes must be optimized to improve the SPEA's performance. Parameters to be optimized include the electrode width and pitch, the gap between the stator and the slider, and the thickness and the material of insulators. In the optimization, the two factors omitted in this study: dynamics and friction, should be considered in addition to the thrust force. Because the electrode structure of the SPEA is complex, such optimization will require computational simulation that cooperate with CAD system, which are capable of calculating electric field in 3-D; for this challenging task, techniques established for MEMS actuators [21] should be introduced.

Second, the actuator needs better guiding. The current means of gap maintenance and lubrication: the glass beads, are sufficient in terms of their function but are not sustainable. The long-stroke planar motion gradually pushes them out of the stator plane. Other guiding methods need to be considered, which include air bearings and coating/embossing of the film surfaces with solid lubricators.

Third, the cables on the slider need to be removed. The SPEA by its nature requires direct feeding of power to its slider, and some applications may not tolerate the obtrusiveness of the power cables. For these applications, one recommendation is to

Authorized licensed use limited to: Southern University of Science and Technology. Downloaded on September 15,2020 at 00:41:38 UTC from IEEE Xplore. Restrictions apply.

use only the stator of the SPEA and use a dielectric sheet as a slider, just like in the original transparent electrostatic actuator in [13]. This abandons the synchronous property of the actuator and weakens the thrust force, but realizes a cable-free slider. Besides this solution, electrostatic actuators of different principles should be considered as well, for their possible application as transparent planar actuators. Potential candidates include electret motors [22], [23], variable capacitance motors [24], [25], corona motors [26], and electrostatic induction motors [27], [28].

Finally, and most importantly, reliability and safety of the actuator must be tested, because the actuator works with high voltages. For human–computer interface applications, even the slightest possibility of accidents must be eliminated. Insulation of the actuator must be tested with strict standards to assure that there will be no discharges, and the driver electronics must be designed to have dependable fail-safe circuits.

VII. Conclusion

A transparent SPEA and its analysis concerning a geometry of electrode arrangement was reported in this paper. A prototype was designed and fabricated based on the analysis, which proved its capability for long-stroke planar motion in experiments. The proposed analytical method allows the future developers of this actuator to invent new electrode arrangement, which should bring improvements to the actuator's performance. We wish that further studies will push this actuator toward its utilization in practical interactive applications.

References

[1] G. Pangaro, D. Maynes-Aminzade, and H. Ishii, "The actuated workbench: Computer-controlled actuation in tabletop tangible interfaces," in *Proc. 15th Annu. ACM Symp. User Interface Softw. Technol.*, 2002, pp. 181–190.

[2] M. Weiss, F. Schwarz, S. Jakubowski, and J. Borchers, "Madgets: Actuating widgets on interactive tabletops," in *Proc. 23rd Annu. ACM Symp. User Interface Softw. Technol.*, 2010, pp. 293–302.

[3] S. Yoshida, H. Noma, and K. Hosaka, "Proactive desk II: Development of a new multi-object haptic display using a linear induction motor," in *Proc. IEEE Virtual Reality Conf.*, 2006, pp. 269–272.

[4] M. Sugi, M. Nikaido, Y. Tamura, J. Ota, T. Arai, K. Kotani, K. Takamasu, S. Shin, H. Suzuki, and Y. Sato, "Motion control of self-moving trays for human supporting production cell "Attentive workbench"," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2005, pp. 4080–4085.

[5] J. de Boeij, E. Lomonova, and A. Vandenput, "Modeling ironless permanent-magnet planar actuator structures," *IEEE Trans. Magn.*, vol. 42, no. 8, pp. 2009–2016, Aug. 2006.

[6] J. W. Jansen, C. M. M. van Lierop, E. A. Lomonova, and A. J. A. Vandenput, "Magnetically levitated planar actuator with moving magnets," *IEEE Trans. Ind. Appl.*, vol. 44, no. 4, pp. 1108–15, Jul/Aug. 2008.

[7] W. Min, M. Zhang, Y. Zhu, B. Chen, G. Duan, J. Hu, and Y. Wensheng, "Analysis and optimization of a new 2-D magnet array for planar motor," *IEEE Trans. Magn.*, vol. 46, no. 5, pp. 1167–71, May 2010.

[8] T. Aoki, K. Asano, R. Ayukawa, H. Ichikawa, Y. Iio, T. Kawase, T. Kuriyama, I. Matsumura, T. Matsushita, H. Mitake, T. Toyama, S. Hasegawa, and M. Sato, "Kobito: Virtual brownies," in *Proc. SIGGRAPH Emerging Technol.*, 2005.

[9] P. Berkelman, M. Miyasaka, and J. Anderson, "Co-located 3D graphic and haptic display using electromagnetic levitation," in *Proc. Haptics Symp.*, 2012, pp. 77–81.

[10] M. Marshall, T. Carter, J. Alexander, and S. Subramanian, "Ultra-tangibles: Creating movable tangible objects on interactive tables," in *Proc. SIGCHI Conf. Human Factors Comput. Syst.*, 2012, pp. 2185–2188.

[11] M. Takasaki, H. Kotani, T. Mizuno, and T. Nara, "Two-dimensional active type surface acoustic wave tactile display on a computer screen," in *Proc. Haptic Interfaces Virtual Environ. Teleoperator Syst.*, 2006, pp. 49–54.

[12] A. Tamura, J. Murayama, Y. Hirata, M. Sato, and T. Harada, "Development of haptic device for touch panel—SPIDAR-tablet—And its haptic computational methods," *J. Virtual Reality Soc. Jpn.*, vol. 16, no. 3, pp. 363–366, 2011.

[13] S. Egawa, T. Niino, and T. Higuchi, "Film actuators: Planar, electrostatic surface-drive actuators," in *Proc. IEEE Micro Electro Mech. Syst. Workshop*, 1991, pp. 9–14.

[14] K. Amano and A. Yamamoto, "Tangible interactions on a flat panel display using actuated paper sheets," in *Proc. ACM Int. Conf. Interactive Tabletops Surfaces*, 2012, pp. 351–354.

[15] T. Hosobata, A. Yamamoto, and T. Higuchi, "2-DOF synchronous electrostatic actuator with transparent electrodes arranged in checkerboard patterns," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2013, pp. 4919–4924.

[16] T. Niino, T. Higuchi, and S. Egawa, "Dual excitation multiphase electrostatic drive," in *Proc. IEEE Ind. Appl. Conf.*, 1995, pp. 1318–1325.

[17] T. Hosobata and A. Yamamoto, "Mixed reality system on flat panel display with real object driven by synchronous transparent electrostatic actuator," in *Proc. Int. Conf. Multimedia Human Comput. Interaction*, 2013, pp. 127-1–127-7.

[18] A. Yamamoto, T. Niino, and T. Higuchi, "Modeling and identification of an electrostatic motor," *Precision Eng.*, vol. 30, no. 1, pp. 104–113, 2006.

[19] A. Yamamoto, T. Niino, T. Ban, and T. Higuchi, "High-power electrostatic motor using skewed electrodes," *Electr. Eng. Jpn.*, vol. 125, no. 3, pp. 50–58, 1998.

[20] T. Niino, A. Yamamoto, and T. Higuchi, "Operation of a dual excitation multiphase electrostatic drive by amplitude-modulated AC voltage," *Electr. Eng. Jpn.*, vol. 131, no. 4, pp. 78–84, 2000.

[21] S. Wiak and K. Smolka, "Integrated computer models of 3-D comb drive electrostatic MEMS structures," *Comput. Methods Innovative Des. Electr. Devices*, vol. 327, pp. 1–18, 2011.

[22] Y. Tada, "Improvement of conventional electret motors," *IEEE Trans. Electr. Insul.*, vol. 28, no. 3, pp. 402–10, Jun. 1993.

[23] T. Genda, S. Tanaka, and M. Esashi, "High power electrostatic motor and generator using electrets," in *Proc. 12th Int. Conf. TRANSDUCERS Solid-State Sens., Actuators, Microsyst.*, 2003, pp. 492–495.

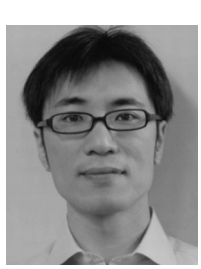
[24] N. Ghalichechian, A. Modafe, J. H. Lang, and R. Ghodssi, "Dynamic characterization of a linear electrostatic micromotor supported on microball bearings," *Sens. Actuators A: Phys.*, vol. 136, no. 2, pp. 496–503, 2007.

[25] N. Yamashita, Z. G. Zhang, A. Yamamoto, M. Gondo, and T. Higuchi, "Voltage-induction type electrostatic film motor driven by two-to four-phase AC voltage and electrostatic induction," *Sens. Actuators A: Phys.*, vol. 140, no. 2, pp. 239–50, 2007.

[26] P. T. Krein, "Analysis of corona motors and micromotors by means of effective gap conductivity," *IEEE. Trans. Ind. Appl.*, vol. 31, no. 4, pp. 752–760, Jul./Aug. 1995.

[27] J. F. Charpentier, Y. Lefevre, E. Sarraute, and B. Trannoy, "Synthesis and modelling of an electrostatic induction motor," *IEEE Trans. Magn.*, vol. 31, no. 3, pp. 1404–1407, May 1995.

[28] S. F. Nagle, C. Livermore, L. G. Frechette, R. Ghodssi, and J. H. Lang, "An electric induction micromotor," *J. Microelectromech. Syst.*, vol. 14, no. 5, pp. 1127–43, 2005.



Takuya Hosobata (S'09–M'12) received the B.S. and M.S. degrees in mechanical engineering from Waseda University, Tokyo, Japan, in 2002 and 2004, respectively. He received the Ph.D. degree in precision engineering from the University of Tokyo, Tokyo, in 2011.

He worked at Sumitomo Heavy Industries Ltd., from 2004 to 2008. He is currently working as a Postdoctoral Researcher. His research interests include electrostatic actuators and mechatronics.

Authorized licensed use limited to: Southern University of Science and Technology. Downloaded on September 15,2020 at 00:41:38 UTC from IEEE Xplore. Restrictions apply.



Akio Yamamoto (S'96–A'99–M'03) received the B.S., M.S., and Ph.D. degrees in precision engineering from the University of Tokyo, Tokyo, Japan, in 1994, 1996, and 1999, respectively.

He was a Research Associate in the Department of Precision Engineering, University of Tokyo, from April 1999 to March 2000, and a Lecturer from April 2000 to March 2005. Since April 2005, he has been an Associate Professor in the same department. His research interests include mechatronics, electrostatic actuators, and human–machine interactions.



Toshiro Higuchi (M'87) received the B.S., M.S., and Ph.D. degrees in precision machinery engineering from the University of Tokyo, Tokyo, Japan, in 1972, 1974, and 1977, respectively.

He was a Lecturer at the Institute of Industrial Science, University of Tokyo, from April 1977 to March 1978, and an Associate Professor from April 1978 to November 1991. Since November 1991, he has been a Professor in the Department of Precision Engineering, University of Tokyo. He was the Leader of the Higuchi Ultimate Mechatronics Project, Kanagawa Academy of Science and Technology, from April 1992 to March 1997. His research interests include mechatronics, magnetic bearings, electrostatic actuators, MEMS, robotics, and manufacturing.

Authorized licensed use limited to: Southern University of Science and Technology. Downloaded on September 15,2020 at 00:41:38 UTC from IEEE Xplore. Restrictions apply.